

Logo Detection App

Project Analysis Report

Generated: 26 March 2026

Project Overview

A semi-automated **logo detection pipeline** for brand advertisement videos. It combines audio transcription, object detection, image filtering, and brand recognition using state-of-the-art models: OWLv2, CLIP, Whisper, Qwen, and LLaVA.

Pipeline Flow

- Video Input
 - Whisper (audio transcription)
 - Qwen LLM (brand name extraction from audio)
 - SceneDetect (keyframe extraction)
 - OWLv2 (logo detection → crops)
 - Filtering (CLIP + heuristics + LLM filters)
 - Brand Recognition Chain (CLIP → OCR/Fuzzy → FAISS DB)
 - Post-filter (Qwen verification)
 - Gradio UI (results output)

Root Level Files

File	Purpose	Status
gradio_app.py	Main Gradio UI entry point	Active
test.py	Standalone OWLv2 video testing script	Dev leftover
requirements.txt	Python dependencies (155 lines)	Active
README.md	Project documentation	Active
repomix-output-*.md	Auto-generated repo dump by Repomix tool	Garbage — delete
.venv/ + .venv311/	Two Python virtual envs (~1.3 GB each, 2.6 GB total)	Should not be in project

/app/ — UI Applications

Three versioned copies of the same app were left in the folder instead of using git history.

File	Purpose	Status
streamlit_faiss_app.py	Main current Streamlit app (1,283 lines)	Active
streamlit_faiss_app_19_10_25.py	Older version of the above	Garbage — duplicate
streamlit_faiss_app_18_10_25.py	Even older version	Garbage — duplicate
streamlit_faiss_app_old.py	Oldest version	Garbage — duplicate
logo_app.py	Legacy Gradio app for testing description models	Unused legacy
app_image.py	Legacy single-image processing app	Unused legacy

/models/ — ML Model Wrappers

File	Purpose	Dead Code
detector_model.py	OWLv2 object detector for logo detection	~165 lines of commented-out old LogoDetector classes
description_model.py	Vision-language models (Gemma, Florence-2, Qwen, BLIP2)	~14 lines of old code + main() test fn with hardcoded Windows paths
faiss_db.py	FAISS vector DB for logo similarity search	Clean
audio_model.py	Whisper audio transcription	~65 lines of commented old WhisperModel implementation
verification_model.py	LLaVA/PaliGemma verification	Clean

/filtering/ — Post-Detection Filters

File	Purpose	Status
clip_filtering.py	CLIP-based false positive removal with clustering	Active
heuristic_filtering.py	Texture, edge, colour, aspect ratio checks	Active
post_filtering.py	Qwen-based correctness verification	Active
llm_filtering.py	LLaMA-based filtering	Uses outdated LLaMAModel — likely broken

/pipelines/ — Processing Orchestration

File	Purpose	Notes
pipeline.py	Main orchestrator: audio→shot detection→logo detection→filtering→brand recognition	Active, 1,007 lines
brand_recognition_chain.py	Strategy pattern chain: CLIP → OCR+Fuzzy → FAISS DB lookup	Active, 953 lines

/utils/ — Utilities

File	Purpose	Status
shot_detection.py	Keyframe extraction using SceneDetect	Active
fuzzy_matching.py	OCR + fuzzy string matching for brand names	Active
prompts.py	System/user prompts for Qwen brand extraction	Active
video_annotations_to_crops.py	Label Studio annotation processor	Legacy — likely unused

Garbage / Dead Code Summary

Category	Files / Lines	Action
Duplicate Streamlit apps	3 files (~3,051 lines)	Delete
Commented-out dead code	~314 lines across 3 model files	Clean up
Auto-generated Repomix file	1 file (3,824 lines)	Delete
Legacy unused apps	2 files (180 lines)	Delete

Category	Files / Lines	Action
Unused utility	video_annotations_to_crops.py	Likely delete
Duplicate virtual envs	.venv/ + .venv311/ (~2.6 GB total)	Remove both, keep one

Estimated savings: ~7,000+ lines of dead/duplicate code removed, ~2.6 GB disk space freed.